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Batch: A2 **Roll No.:** 16010324044
Experiment / assignment / tutorial No. 1
Grade: AA / AB / BB / BC / CC / CD /DD
Signature of the Staff In-charge with date

TITLE: Introduction to packet tracer and networking devices

AIM: To configure the given network in CISCO packet tracer using various networking devices

OUTCOME: Student will be able to understand layered architecture of communication network and physical layer services

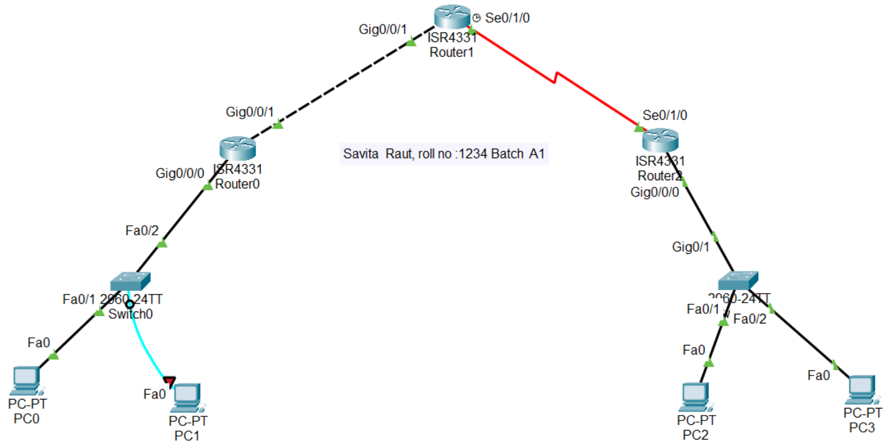
Theory:
Packet Tracer is a cross-platform visual simulation tool designed by Cisco Systems that allows users to create network topologies and imitate modern computer networks. The software allows users to simulate the configuration of Cisco routers and switches using a simulated command line interface. Packet Tracer makes use of a drag and drop user interface, allowing users to add and remove simulated network devices as they see fit.

Different networking devices are available in packet tracer. End devices, switches, routers, Access points, Hubs, etc. The devices can be used to designed different network topologies as per the user requirement. The combination may contain a router, a switch and end devices such as laptop, printer, PC, etc.

Two similar devices can be connected by using a cross over cable. Two different devices can be connected by straight through cables. If the device has optical fiber port then optical fiber cables can be used for communication. Cat 5 or Cat 6e are the most widely used cables for communication between the wired devise using RJ 45 connector. Console cable can be used for programming the devices. Connection between the devices can be verified in different ways. PDUs are available to check the same.
Different servers can be configured using cable modems, cloud and server.

Procedure:

1. To begin with the experiment install the latest version of packet tracer
2. Using the packet tracer software configure the following topology

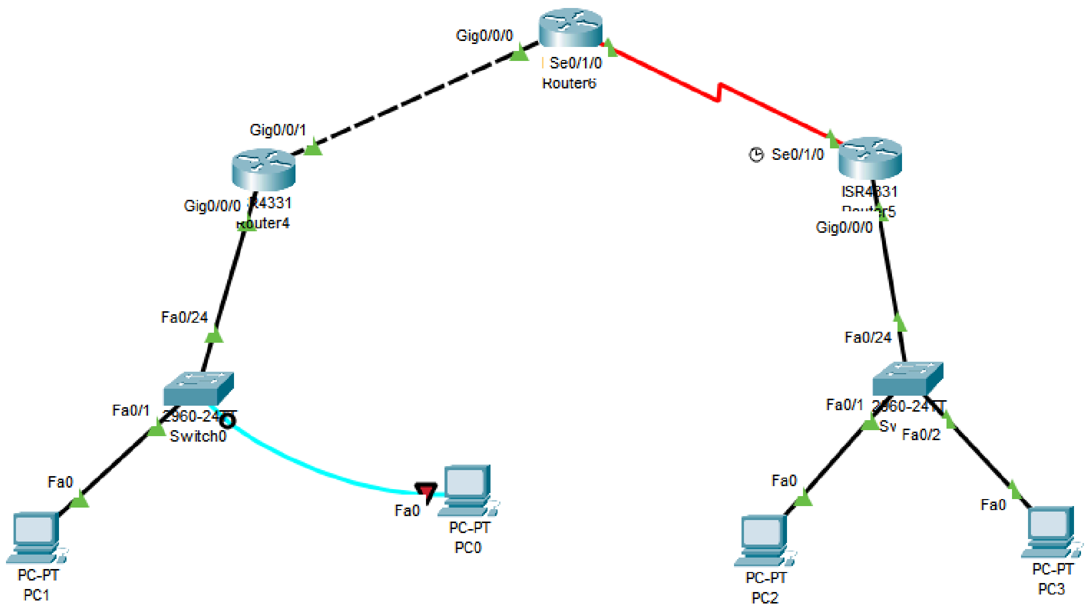


3. For the configuration use console cable between the PC0 and switch 0.
4. Use 2960 switch for switch 0 and switch 1
5. Use 4331 router and add the serial ports to the routers use NIM-2T module to insert serial ports in the router
6. 4331 router is used to have user defined network wherein multiple slots are available to add modules.
7. Connect PC to switch using fast Ethernet port
8. Connect switch and router using fast Ethernet on switch and gigabit Ethernet port on the router.
9. Similar configuration has to be done on the other side.
10. Once the configuration is done all the ports needs to be up
11. At the end all the connections should be up.

Observations:

Image when all ports are in setup phase or DOWN state.

Image of the topology when all the ports are ON.



Explain in details following Network devices

- **Router**

A router is a Layer 3 networking device that forwards packets between different networks using IP addresses. It determines the best path for data transmission using routing tables and routing protocols. Routers connect LANs to WANs and provide communication between different networks.

- **Switch**

A switch is a Layer 2 networking device used to connect devices within the same Local Area Network (LAN). It forwards frames using MAC addresses and creates separate collision domains for each port, improving network efficiency.

- **Hub**

A hub is a Layer 1 device that simply repeats incoming signals to every connected port without examining the data. It does not maintain any address table, making it inefficient compared to switches.

- **Bridge**

A bridge is a Layer 2 device used to connect two LAN segments. It filters frames based on MAC addresses and reduces unnecessary traffic by forwarding frames only to the required network segment.

CONCLUSION

In this experiment, I learnt how to use Cisco Packet Tracer. Various types of networking devices such as routers, switches and PCs were configured to create the required topology. Different types of network cables were used for interconnection, and all interfaces were brought to the UP state successfully. This experiment helped in understanding basic networking devices, cable selection, network topology creation and the working of Packet Tracer.

Signature of faculty in-charge